

Expt-5

AIM: Separation of binary organic mixture by acid-base extraction and analysis using given FTIR and NMR spectrum.

THEORY:

The transfer of a solute from one phase to another is a very common technique in organic chemistry. It is called extraction, and it is also a common technique in everyday life. Extraction involves dissolving a compound or compounds either (1) from a solid into a solvent or (2) from a solution into another solvent.

Some organic molecules possess acidic or basic functional groups. These groups can be protonated or deprotonated in a solution given the right pH. When the organic molecule has undergone an acid-base reaction it becomes charged in the form of an ionic salt. In turn, this charge increases the solubility of the molecule in aqueous (or polar) solvents, and decreases the solubility of the molecule in nonpolar, organic/greasy solvents. This solubility difference allows the molecule to be selectively pulled out of the “soup” of other molecules. It might seem particularly daunting to try and selectively remove all of one molecule from a plant or animal while leaving the millions of other types of molecules behind. Nevertheless, extracting molecules from natural sources is a fairly common procedure and can be done on an industrial scale for a variety of biologically active molecules. Indeed, many important life-saving drugs on the market are originally isolated from natural sources.

When infrared light is passed through a sample of an organic compound, some of the frequencies are absorbed and others are transmitted. The plot of absorbance or transmittance against the frequency represents the infrared spectrum. However, the convention is the absorbance or transmittance is plotted against the wavenumber (increasing order from right to left) to represent the Fourier transform infrared (FTIR) spectrum. Different types of chemical bonds (like C-C, C=C, C-N, C-O, C=O, C=N, C-H, N-H etc.) in organic compound have different vibrational frequencies, and the presence of these bonds can be detected by identifying their characteristics frequencies as absorption bands in infrared spectrum.

EQUIPMENT AND MATERIALS.

Mixture of 500 mg each of benzoic acid and nitroaniline, HCl, NaOH, 2 Beakers 100 ml, 2 Conical Flask 100 ml, funnel, glass rod, 2 FTIR plots.

PROCEDURE:

1. Slowly adding 6 M NaOH and Stir the solution frequently. Remember that the neutralization reaction of acids and bases is exothermic, so the flask should still be in an ice bath.
2. Check the pH with pH paper to make sure it's basic, adding more base if necessary.
3. When a precipitate forms in your extract, collect it by vacuum (suction) filtration on a piece of filter paper in a Büchner funnel.
4. Make the NaOH extract acidic with 6 M HCl, following the procedure outlined above. You should see a precipitate when the extract has been acidified.
5. When a precipitate forms in your extract, collect it by vacuum (suction) filtration on a piece of filter paper in a Büchner funnel.
6. Both the precipitate was dried well in a vacuum thermal chamber at 80 °C.
7. Record the melting point and explain the given FTIR spectra for the samples.

OBSERVATION:

1. Physical characteristics (Fill it according to the physical properties of the given mixture):

- a. State:
- b. Texture:
- c. Color:
- d. Order:

Compound	Weight taken (g)	Weight recovered (g)	Melting point measured (°C)	Melting point reported (°C)
Benzoic acid	0.5			
Nitro aniline	0.5			

FTIR ANALYSIS:

(Draw the structure of the compounds)

FTIR of Benzoic acid

(Explain the given spectrum in terms of delocalized ArC=C, ArC-H, carboxylic acid COO, hydrogen bonded acidic-OH)

FTIR of Nitro aniline

(Explain the given spectrum in terms of delocalized ArC=C, ArC-H, ArC-N, NO₂, hydrogen bonded acidic-NH₂)

CALCULATION:

Calculate the % recovery for Benzoic acid, Nitro aniline and for total mixture using the following equation;

$$\% \text{ Recovery} = \frac{\text{Weight recovered}}{\text{Weight taken}} \times 100$$

RESULT:

The % recovery for Benzoic acid with observed melting point of

The % recovery for Nitro aniline with observed melting point of

The % recovery for Total mixture

PRECAUTIONS:

Write 3/4 points about the precautions that you have taken during your experiment.
